

SHP2 - Target Landscape Report

Target Protein: SHP2

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1. Target Profile

Protein Overview

SHP2 is a protein with the gene name PTPN11. It has an intron-containing transcript variant, indicating it is a multi-exonic protein.

Molecular Function & Mechanism

SHP2 functions as a dual-specificity phosphatase, involved in dephosphorylating both tyrosine and serine/threonine residues on target proteins. This activity plays a crucial role in modulating various signaling pathways.

Signalling Pathways

SHP2 participates in multiple signalling pathways including the Ras signaling pathway, Phospholipase D signaling pathway, Efferocytosis, Axon guidance, IgSF CAM signaling, C-type lectin receptor signaling pathway, JAK-STAT signaling pathway, Natural killer cell mediated cytotoxicity, and T cell receptor signaling pathway. Its role in these pathways is significant for cellular processes such as growth, differentiation, and immune responses.

Therapeutic Rationale

SHP2 is a compelling drug target due to its involvement in multiple critical signalling pathways that are relevant to various diseases. It is implicated in conditions such as Noonan syndrome and LEOPARD syndrome, which have strong evidence based on Open Targets scores. Additionally, SHP2's role in immune responses and cell growth makes it challenging but also highly interesting for therapeutic intervention.

Sources: UniProt * Open Targets * KEGG

2. Clinical Trials

Source: [ClinicalTrials.gov](https://clinicaltrials.gov)

46

Total Studies

14

Completed

11

Ongoing / Active

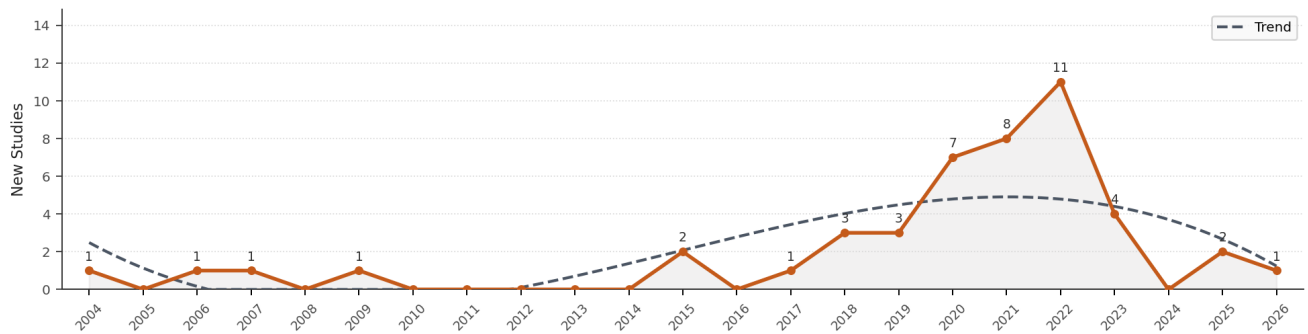
2

Withdrawn

19

Other

Clinical Trials Started per Year



Clinical trials started per year with polynomial trend line.

Associated Conditions / Diseases

Condition	Studies
Colorectal Cancer	8
Non-small Cell Lung Cancer	7
Advanced Solid Tumor	6
Solid Tumor	5
Head and Neck Squamous Cell Carcinoma	3
Pancreatic Ductal Carcinoma	3
Non-Small Cell Lung Cancer	3
Non Small Cell Lung Cancer	3
Pancreatic Cancer	3
Esophageal Squamous Cell Carcinoma	2
Breast Cancer	2

Condition	Studies
Other Solid Tumors	2
Advanced Solid Tumors	2
Advanced Cancer	2
Solid Tumors	2

All Interventions (unique entries from trial data)

All unique intervention names recorded across clinical trials for this target, with the NCT ID(s) of the studies they appear in.

Intervention Name	NCT ID(s)
BBP-398	NCT05480865, NCT05621525
BBP-398 with nivolumab	NCT05375084
Gilteritinib	NCT05279859
Sotorasib	NCT04959981
TNO155	NCT05541159

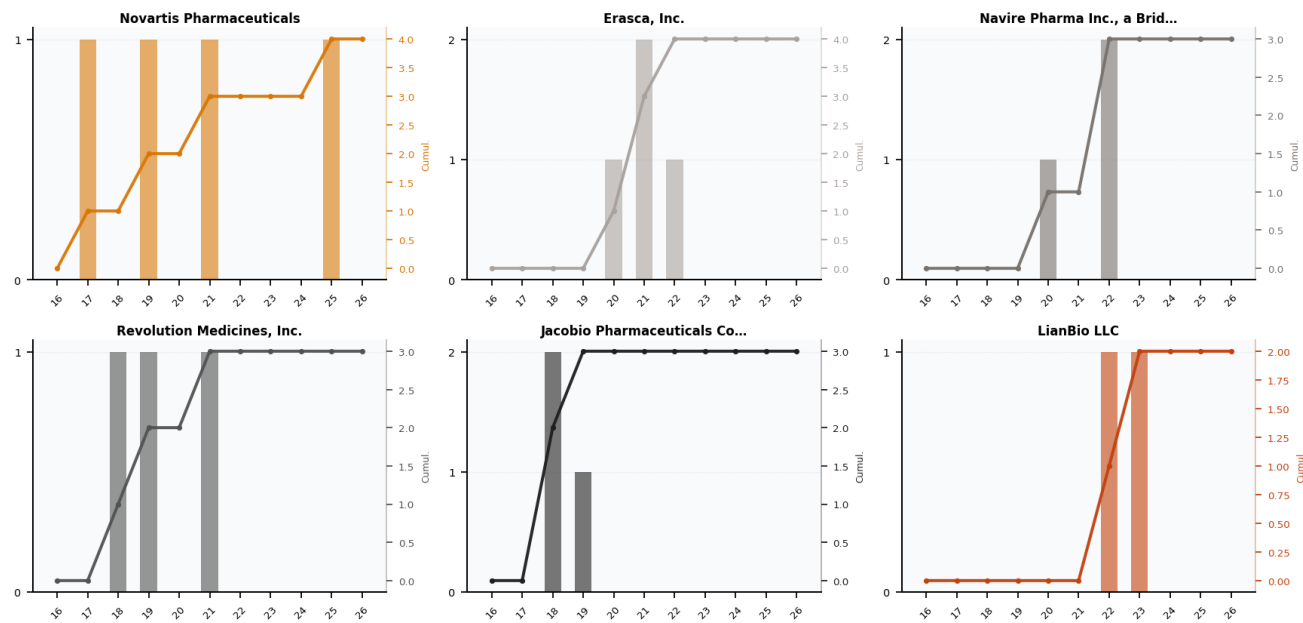
Combination Therapies / Products

Studies where two or more interventions passed the drug-name filter. Each row represents one study.

Interventions	NCT ID
JDQ443 + TNO155	NCT04699188
PF-07284892 + binimetinib + cetuximab + encorafenib + lorlatinib	NCT04800822
LY3214996 + RMC-4630	NCT04916236
RMC-4630 + Sotorasib	NCT05054725
JAB-21822 + JAB-3312	NCT05288205
MK-0472 + Pembrolizumab	NCT05853367
BBP-398 + osimertinib	NCT06032936

Lead Sponsors - Activity & Key Studies

Lead Sponsors — Studies per Year (last 10 yrs)



Lead sponsors - annual count (bars) and cumulative total (line).

Bars = annual study count * Line = cumulative total * Last 10 years.

Novartis Pharmaceuticals

Study Title	NCT ID	Phase	Status	Start	End
Pharmacokinetics Study of TNO155 in Participants With Mild, Moderate, or Se...	NCT05541159	PHASE1	WITHDRAWN	2025-03-19	2025-07-15
Study of JDQ443 in Patients With Advanced Solid Tumors Harboring the KRAS G...	NCT04699188	PHASE1 / PHASE2	ACTIVE_NOT_RECRUITING	2021-02-24	2027-01-25
Phase Ib Study of TNO155 in Combination With Spartalizumab or Ribociclib in...	NCT04000529	PHASE1	TERMINATED	2019-07-30	2024-01-15

Study Title	NCT ID	Phase	Status	Start	End
Dose Finding Study of TNO155 in Adult Patients With Advanced Solid Tumors	NCT03114319	PHASE1	TERMINATED	2017-05-26	2025-07-04

Erasca, Inc.

Study Title	NCT ID	Phase	Status	Start	End
A Study of Anti-Cancer Therapies Targeting the MAPK Pathway in Patients Wit...	NCT05279859	PHASE1 / PHASE2	WITHDRAWN	2022-03-15	2025-06-01
A Study of Anti-Cancer Therapies Targeting the MAPK Pathway in Patients Wit...	NCT04959981	PHASE1	COMPLETED	2021-09-02	2023-04-27
A Study of ERAS-007 as Monotherapy or in Combination With ERAS-601 in Patie...	NCT04866134	PHASE1 / PHASE2	ACTIVE_NOT_RECRUITING	2021-05-07	2025-11-01
A Dose Escalation /Expansion Study of ERAS-601 in Patients With Advanced or ...	NCT04670679	PHASE1	ACTIVE_NOT_RECRUITING	2020-12-15	2026-06

Navire Pharma Inc., a BridgeBio company

Study Title	NCT ID	Phase	Status	Start	End
SHP2 Inhibitor BBP-398 in Combination With Nivolumab in Patients With Advan...	NCT05375084	PHASE1	TERMINATED	2022-10-20	2024-07-15

Study Title	NCT ID	Phase	Status	Start	End
SHP2 Inhibitor BBP-398 in Combination With Sotorasib in Patients With Advan...	NCT05480865	PHASE1	TERMINATED	2022-07-06	2024-08-22
First-in-Human Study of the SHP2 Inhibitor BBP-398 in Patients With Advance...	NCT04528836	PHASE1	TERMINATED	2020-11-12	2024-07-30

Revolution Medicines, Inc.

Study Title	NCT ID	Phase	Status	Start	End
Combination Study of RMC-4630 and Sotorasib for NSCLC Subjects With KRASG12...	NCT05054725	PHASE2	COMPLETED	2021-12-30	2024-08-29
Dose-Esc/Exp RMC4630 & Cobi in Relapsed/Refractory Solid Tumors & RMC4630& ...	NCT03989115	PHASE1 / PHASE2	COMPLETED	2019-07-02	2022-02-08
Dose Escalation of RMC-4630 Monotherapy in Relapsed/Refractory Solid Tumors	NCT03634982	PHASE1	UNKNOWN	2018-09-28	2023-05-31

Jacobio Pharmaceuticals Co., Ltd.

Study Title	NCT ID	Phase	Status	Start	End
A First-in-Human, Phase 1 Study of JAB-3312 in Adult Patients With Advanced...	NCT04045496	PHASE1	COMPLETED	2019-09-26	2022-12-16

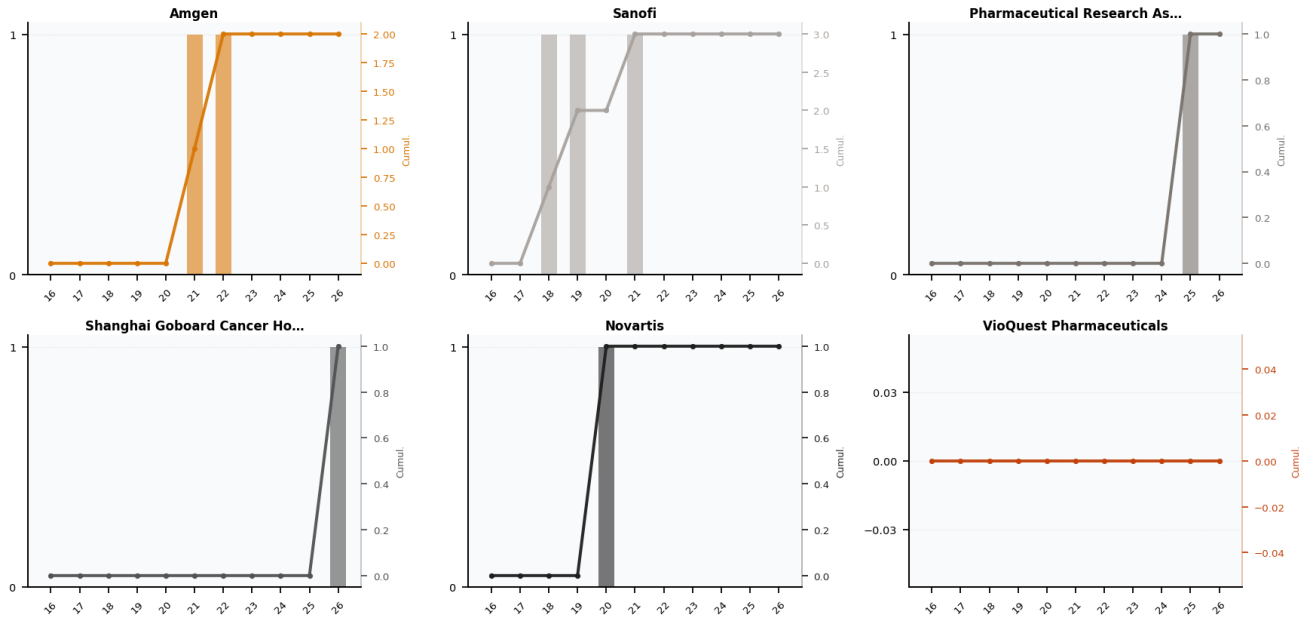
Study Title	NCT ID	Phase	Status	Start	End
A First-in-Human Study of JAB-3068 (SHP2 Inhibitor) in Adult Patients With ...	NCT03565003	PHASE1 / PHASE2	COMPLETED	2018-11-20	2023-01-31
A First in Human, Dose Escalation Study of JAB-3068 (SHP2 Inhibitor) in Adu...	NCT03518554	PHASE1	COMPLETED	2018-04-23	2022-02-03

LianBio LLC

Study Title	NCT ID	Phase	Status	Start	End
BBP-398 in Combination With Osimertinib in Locally Advanced or Metastatic N...	NCT06032936	PHASE1	TERMINATED	2023-07-27	2024-03-29
Phase I Study of the BBP-398 in Patients With Advance Solid Tumors	NCT05621525	PHASE1	TERMINATED	2022-10-18	2024-03-29

Collaborating Institutions - Activity & Key Studies

Collaborating Institutions — Studies per Year (last 10 yrs)



Collaborating institutions - annual count and cumulative total.

Bars = annual study count * Line = cumulative total * Last 10 years.

Amgen

Study Title	NCT ID	Phase	Status	Start	End
SHP2 Inhibitor BBP-398 in Combination With Sotorasib in Patients With Advan...	NCT05480865	PHASE1	TERMINATED	2022-07-06	2024-08-22
Combination Study of RMC-4630 and Sotorasib for NSCLC Subjects With KRASG12...	NCT05054725	PHASE2	COMPLETED	2021-12-30	2024-08-29

Sanofi

Study Title	NCT ID	Phase	Status	Start	End
Combination Study of RMC-4630 and Sotorasib for NSCLC Subjects With KRASG12...	NCT05054725	PHASE2	COMPLETED	2021-12-30	2024-08-29

Study Title	NCT ID	Phase	Status	Start	End
Dose-Esc/Exp RMC4630 & Cobi in Relapsed/Refra ctory Solid Tumors & RMC4630& ...	NCT03989115	PHASE1 / PHASE2	COMPLETED	2019-07-02	2022-02-08
Dose Escalation of RMC-4630 Monotherapy in R elapsed/Refractor y Solid Tumors	NCT03634982	PHASE1	UNKNOWN	2018-09-28	2023-05-31

Pharmaceutical Research Associates

Study Title	NCT ID	Phase	Status	Start	End
Pharmacokinetics Study of TNO155 in Participants With Mild, Moderate, or Se...	NCT05541159	PHASE1	WITHDRAWN	2025-03-19	2025-07-15

Shanghai Goboard Cancer Hospital

Study Title	NCT ID	Phase	Status	Start	End
A Study to Evaluate Safety, PK and Efficacy of GH55 in Combination With GH2...	NCT07446725	PHASE1 / PHASE2	NOT_YET_RECR UITING	2026-02-24	2029-12-31

Novartis

Study Title	NCT ID	Phase	Status	Start	End
Adagrasib in Combination With TNO155 in Patients With Cancer (KRYSTAL 2)	NCT04330664	PHASE1	COMPLETED	2020-04-07	2025-02-24

VioQuest Pharmaceuticals

No study detail available.

Withdrawn Studies & Reasons

Study Title	NCT ID	Start	Reason
Pharmacokinetics Study of TNO155 in Participants With Mild, Moderate, or Severe ...	NCT05541159	2025-03-19	Sponsor's decision
A Study of Anti-Cancer Therapies Targeting the MAPK Pathway in Patients With Hem...	NCT05279859	2022-03-15	Sponsor decision.

Trial & Patent Data Files

Trials CSV (27 records)	SHP2/csvs/SHP2_trials.csv
Patents CSV (3 records)	SHP2/csvs/SHP2_patents.csv

Sources: *ClinicalTrials.gov* (trials) * *EPO Espacenet* (patents)

3. Drug Discovery Landscape

11

Active Dimensions

2

Emerging Dimensions

13

Total Shown

For a given protein target, this analysis identifies drug-discovery dimensions explored in the literature and evaluates their volume and recency to highlight factors most relevant to the program. It focuses on publications from the past 5 years across key drug-discovery dimensions.

Landscape Summary

Property	Value
Active Dimensions	11
Emerging Dimensions	2
Total Shown	13

Dimensions Detail

Rank	Area	Dimension	Status	PMIDs
1	Mutations & Variants	Resistance Mutations	Active	36747633 36747633 38226159 36747713 41775683
2	Combination Therapy	Combination / Synergy	Active	39253995 40900823 36752207 36780229 36739480
3	Binding Site	Orthosteric / Active Site	Active	32024694 41628498 31789461 36584630 30186742
4	Binding Site	Allosteric Site / Modulator	Active	35127377 36999992 36476046 34936223 27505669
5	Modality	Degraders / PROTACs / Molecular Glues	Active	40609222 39089586 39300128 38171010 36427459
6	Mutations & Variants	Activating / Driver Mutations	Active	41326795 41091575 40128633 35101231 40166154
7	Binding Mode	Covalent / Irreversible	Active	42033018 41561450 38618612 40304209 35404998

Rank	Area	Dimension	Status	PMIDs
8	Preclinical & Clinical	Clinical Candidate / Trial	Active	41484121 41781328 42114409
9	Selectivity Strategy	Selectivity Profiling / Off-Target Panel	Active	38842946 36938364 32864202
10	Selectivity Strategy	Dual / Multi-Target	Active	40381223 35447031
11	Hit Discovery & Optimisation	Fragment-Based / HTS	Active	38462716 35781901
12	Hit Discovery & Optimisation	SAR / Lead Optimisation	Emerging	40808802 28388567 11341829
13	Combination Therapy	Synthetic Lethality / Pathway Co-targeting	Emerging	32001483

Must-Read Cross-Cutting Papers

- [Targeting SHP2 with an Active Site Inhibitor Blocks Signaling and Breast Cancer Cell Pheno...](#)
- [A hotspot phosphorylation site on SHP2 drives oncoprotein activation and drug resistance.](#)
- [A Hotspot Phosphorylation Site on SHP2 Drives Oncoprotein Activation and Drug Resistance.](#)

4. ChEMBL

This section curates approved and investigational small-molecule drugs associated with the target protein from ChEMBL and maps them to corresponding PDB structures. Each molecule is annotated with its highest reported binding activity against the primary target and its selectivity profile relative to off-target proteins identified in ChEMBL bioactivity data.

Molecules

Status

No molecules found.

** Name: preferred drug name. * Phase: highest clinical phase reached (1-4; 4 = approved). * Best pChEMBL: highest $-\log_{10}(\text{molar})$ activity value in ChEMBL against the primary target; [Ref] links to the source document. * Target PDB: PDB entry IDs where the ligand is co-crystallised with the primary target (human / mouse). * Selectivity: difference between on-target pChEMBL and the best off-target pChEMBL (positive = more potent on-target). Top off-target gene and its activity shown in parentheses; [Ref] links to the source document. * Off-Targets: up to 4 off-target gene symbols ranked by pChEMBL whose bioactivity exceeded the potency threshold in ChEMBL.*

ChEMBL Activity Data Processing



This section summarizes molecular activity data reported in ChEMBL across multiple activity metrics. It highlights publication trends over time for the target, quantifies the number of studies/documents associated with each metric, and includes scaffold-level information for molecules reported in each document within the accompanying CSV files.

19

Molecules in

14

Molecules after filter

2

Studies passing

Activity Thresholds & Record Counts

Standard Type	Threshold	Docs (passed)	Mols in docs
Activity	≥ 50	2	14
IC50	$\leq 10,000$ nM	1	4

Activity thresholds are applied to retain only studies whose most potent molecule meets a minimum activity criterion - studies where even the best molecule fails to reach this bar are excluded from downstream analysis. 'Mols in docs' counts all unique molecules across surviving documents for that standard type, not just the single best molecule selected per document.

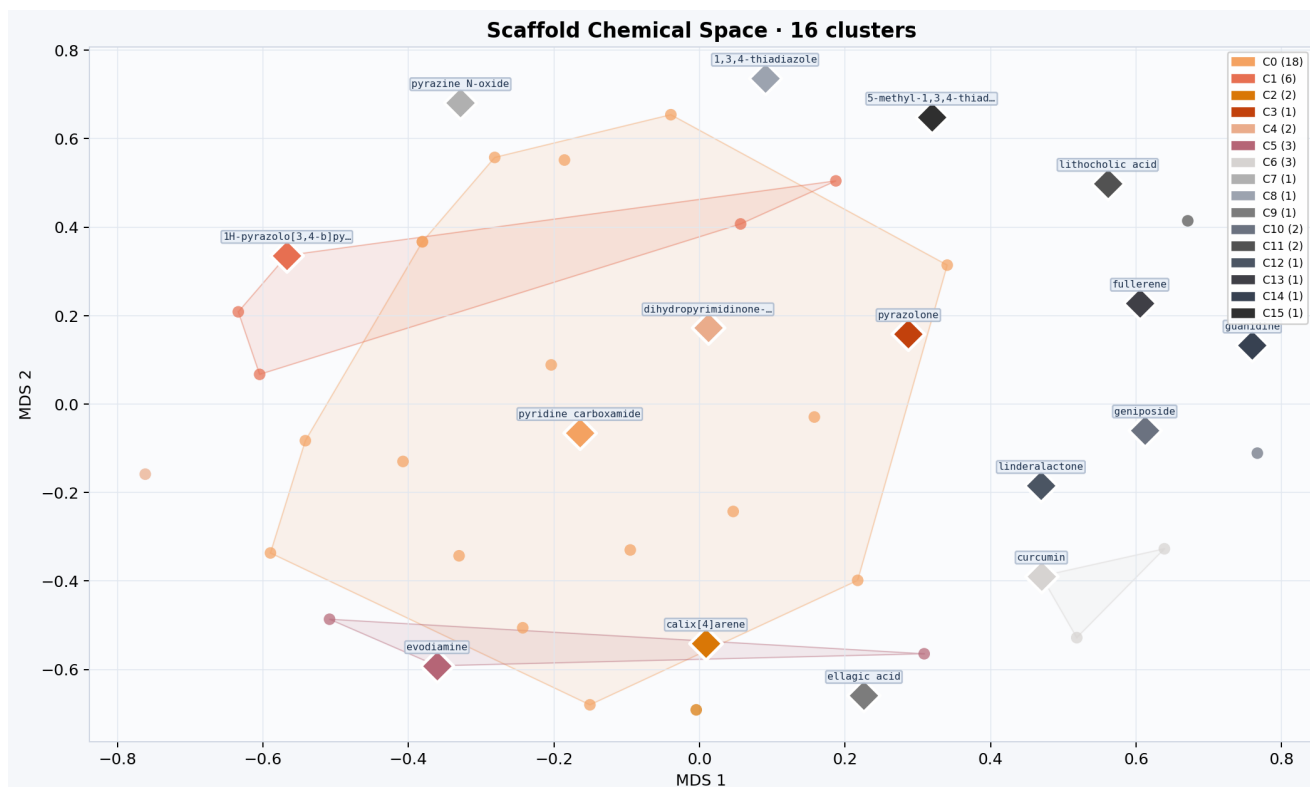
5. Scaffold Study

Scaffold mining uses PubMed literature for the target, screens titles with a local LLM, extracts scaffold names, fetches SMILES from PubChem and CACTUS, then deduplicates by name. The sorted CSV is the primary input for clustering and scaffold hopping.

Scaffolds by Publication Year

Year	Scaffolds
2026	pyrazolo[3,4-d]pyrimidin-4-one, dihydropyrimidinone-pyrazole, quinolone, calix[4]arene
2025	liensinine, benzoazepinamine, imidazo[2,1-b][1,3,4]thiadiazole, pyrazine N-oxide, 5-methyl-1,3,4-thiadiazole-2-thiol, pyrazolopyrimidinone, pyrrolopyrimidine, phenyl urea (+4 more)
2024	curcumin, pyridine carboxamide, pyrazolopyrazine, Pyrido[1,2-a]pyrimidin-4-one, furanyl amide, ellagic acid, lanostane
2023	guanidine, 1,3,4-thiadiazole, linderalactone, phenyl-pyrazolone, lithocholic acid, imidazopyrazine
2022	pyrazoline, aminopyrazine, dihydrotanshinone
2021	triazoloquinazolinone, benzimidazole, pyrimidine
2020	pyrazolone, geniposide, pyridine, 3,4,6-trihydroxy-5-oxo-5H-benzo[7]annulene
2019	evodiamine
2018	(3'-amino-[1,1'-biphenyl]-4-yl) sulfamic acid
2017	benzo[c][1,2,5]thiadiazole, dibenzofuran
2016	phidianidine B
2014	fullerene
2007	retinoid

Scaffold Clustering



Scaffold chemical space coloured by Butina cluster. Diamonds = medoids.

Clustering Parameters

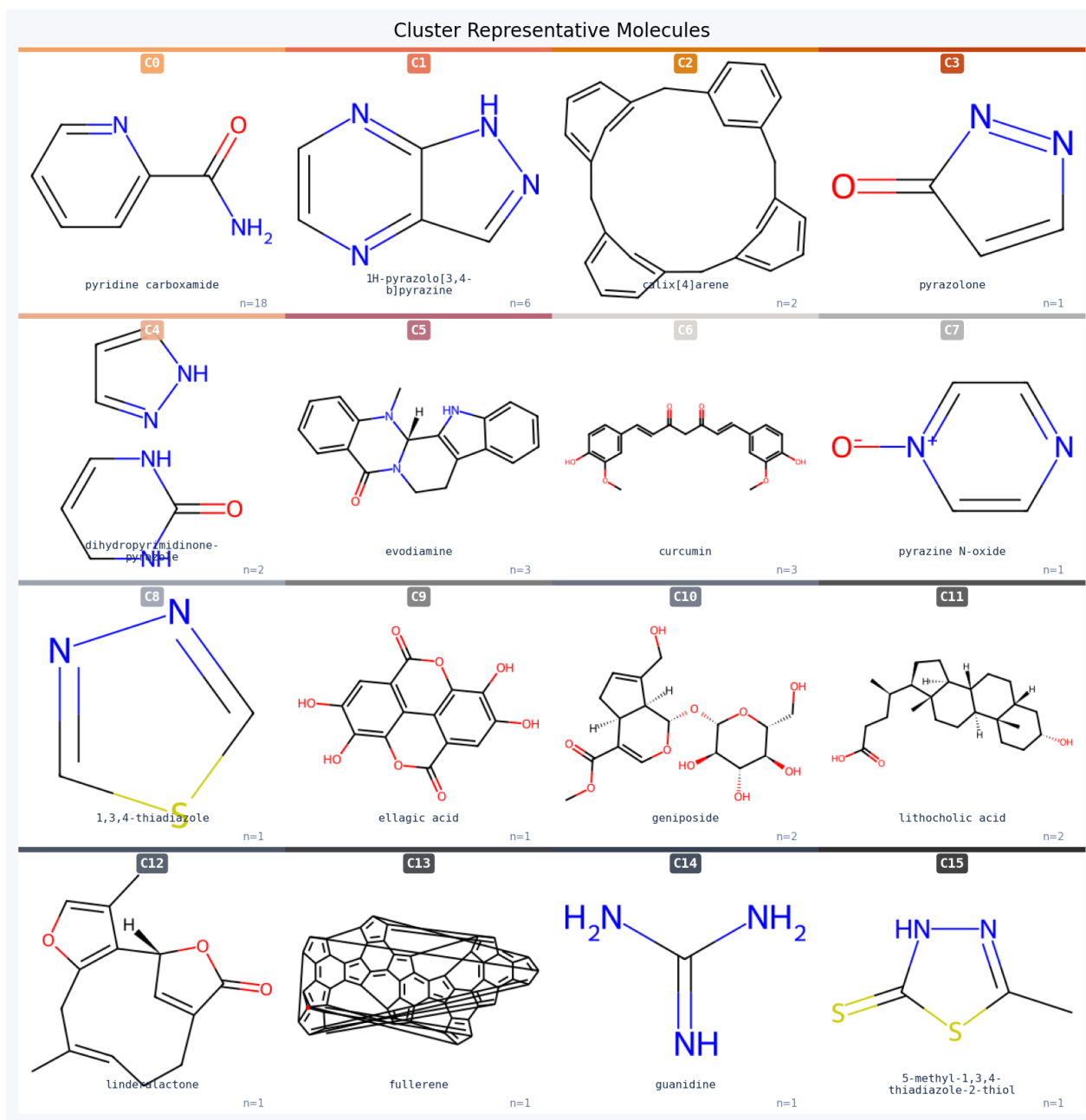
Property	Value
Dimensionality Reduction	MDS
Similarity Method	Butina / Tanimoto
Fingerprint	Morgan r=2, 2048 bits
Similarity Cutoff	0.15
Clusters Found	16
Valid Molecules	46

Selected Central Scaffold

Property	Value
Scaffold Name	pyridine carboxamide
SMILES	<chem>NC(=O)c1ccccc1</chem>
Cluster summary CSV	0 records - SHP2/csvs/SHP2_scaffolds_sorted_cluster_summary.csv

The cluster with the highest number of unique scaffolds was selected as the primary cluster. Its medoid - the molecule with the highest average Tanimoto similarity to all other cluster members - was chosen as the central

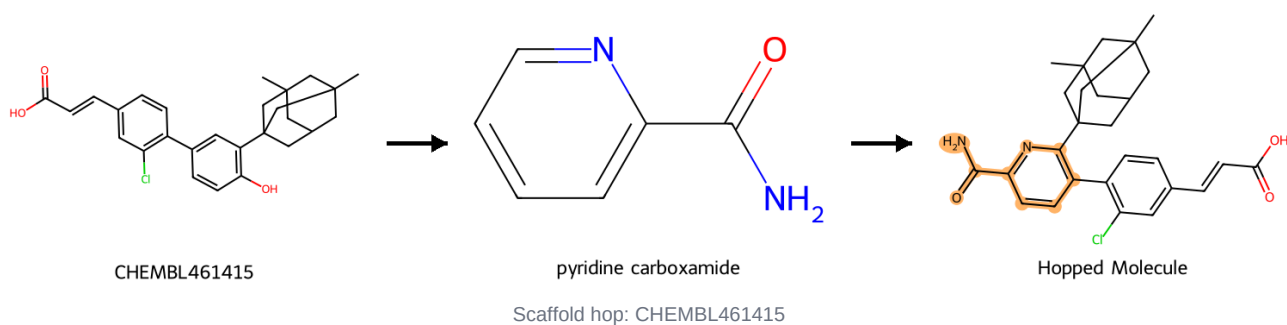
scaffold for downstream scaffold hopping. The chemical space map shows all scaffolds coloured by cluster with convex hulls; the cluster centres panel shows the representative structure per cluster.



Cluster representative scaffolds (medoids).

Scaffold Hopping & Fragment Replacement

The central scaffold identified from the clustering analysis was selected for scaffold hopping. Molecules from the ChEMBL IC50 dataset exhibiting the highest scaffold pharmacophore similarity to the central scaffold, along with selected reference molecules from the approved/investigational compounds list, were used as the basis for the scaffold hopping process.



Fragment Replacement (FARE)

Fragments with physicochemical properties similar to those already present were extracted from the fragment library and attached to the scaffold using the selected molecule as the reference framework.

Generated Molecule Files

CSV File	Path	Molecules Generated
pyridine carboxamide_CHEMBL461415_FARE_generated_molecules.csv	SHP2/generated/pyridine carboxamide_CHEMBL461415_FARE_generated_molecules.csv	2420

Generated Data Files

The following data files were generated during this analysis and are archived for reference and further processing.

File Name	Type	Records	Location
SHP2_trials.csv	CSV	varies	SHP2
SHP2_patents.csv	CSV	varies	SHP2
SHP2_scaffolds_sorted.csv	CSV	varies	SHP2
SHP2_scaffolds_sorted_cluster_summary.csv	CSV	varies	SHP2